

JOSEPH TAN

Senior Full-Stack Engineer

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Professional Summary

Senior Full-Stack Engineer with experience building SaaS platforms, logistics systems, internal business applications, and frontend-heavy operational tools. Strong background across React, Node.js, PostgreSQL, Redis, and AWS environments with focus on backend scalability, frontend performance, production reliability, and maintainable system architecture.

Experienced handling production debugging, deployment coordination, concurrency-related edge cases, and performance bottlenecks across high-volume reporting and synchronization workflows. Prefer practical long-term systems over overly abstract architectures.

Worked with asynchronous processing, Redis caching, background jobs, API reliability improvements, and gradual migration efforts between legacy and newer platform components.

Core Technical Skills

Frontend

React.js, Next.js, Vue.js, TypeScript, Redux Toolkit, Tailwind CSS

Backend

Node.js, Express.js, NestJS, Python, Django, GraphQL, REST APIs

Databases

PostgreSQL, MySQL, MongoDB, Redis, Query Optimization

Infrastructure

AWS EC2, Docker, Linux, Nginx, GitHub Actions, CI/CD

Engineering Focus

Backend Scalability, Production Reliability, API Performance, Async Processing, Caching Strategy

Leadership

Code Review, Mentoring, Sprint Planning, Agile/Scrum

Selected Engineering Achievements

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- Improved system stability for a warehouse management platform supporting 7 regional warehouses by resolving synchronization inconsistencies, duplicate writes, and concurrent update conflicts.
 - Optimized PostgreSQL query behavior and Redis caching during peak traffic periods by restructuring query patterns, reducing lock contention, and separating reporting workloads from transactional operations.
 - Improved frontend performance across internal dashboards by reducing unnecessary re-renders, splitting large bundles, and removing inefficient polling behavior.
 - Participated in a gradual ERP migration effort where legacy and newer services needed to coexist without disrupting downstream integrations and operational processes.
 - Designed asynchronous job handling for reporting exports, bulk imports, and long-running processing tasks to reduce request blocking and improve reliability.

Professional Experience

Freelance Senior Full-Stack Developer

Shanghai, China | May 2023 – Present

Worked with a small engineering team building SaaS platforms, logistics systems, and internal operational software for overseas clients.

Responsibilities & Contributions

- Developed backend APIs, deployment infrastructure, and business-critical tooling for logistics and warehouse platforms.
- Handled deployment preparation, rollback planning, production debugging, and release follow-ups during deployment cycles.
- Improved API reliability under concurrent synchronization workloads involving inventory updates and bulk processing.
- Resolved frontend performance bottlenecks caused by oversized table rendering and repeated API polling behavior.
- Built role and permission management systems with region-based access controls and safer inheritance boundaries.
- Worked with Redis caching and background processing to reduce database contention during peak operational periods.
- Investigated cascading API failures, synchronization delays, and bulk import edge cases during high-volume warehouse cutoff periods.

Full-Stack Developer — QPSoftware

Shanghai, China | Jul 2020 – Mar 2023

Website: <https://www.qpsoftware.com.cn>

Developed CRM platforms and internal business systems for manufacturing and retail clients with operationally different business requirements.

Responsibilities & Contributions

- Developed CRM modules, dashboards, and internal management systems used by manufacturing and retail organizations.
- Gradually migrated legacy jQuery-based interfaces into React.js while maintaining production continuity and minimizing workflow disruption.
- Improved backend endpoint performance by optimizing heavy joins, adjusting indexing strategies, and moving expensive processing into asynchronous jobs.
- Participated in platform migration efforts requiring coexistence between older modules and newer services during staged rollouts.
- Simplified internal workflows and dashboard navigation after repeated feedback from onboarding and operations teams.
- Worked closely with QA teams during release cycles to investigate staging-to-production inconsistencies across client environments.

Education

Bachelor's Degree in Computer Science

Xuzhou University of Technology — China

Sep 2016 – Jun 2020

Additional Information

- Strong interest in backend reliability, distributed synchronization behavior, caching strategy, and production debugging.
- Comfortable handling both software development and infrastructure-side troubleshooting during operational incidents.
- Experienced working on reporting systems, approval workflows, warehouse synchronization flows, and frontend-heavy internal platforms.
- Prefer maintainable long-term systems over overly complex architectures with high operational overhead.
- Interested in concurrency handling, API reliability, observability, failure recovery, and reducing fragile dependencies between services.